Token on Me: A Token-Level Dataset of Company Authorship in FLOSS Codebases

Ellian Carlos Costa
University of São Paulo, IME-USP
São Paulo, Brazil
elliancarlos@ime.usp.br

Arthur Pilone
University of São Paulo, IME-USP
São Paulo, Brazil
arthurpilone@ime.usp.br

David Tadokoro
University of São Paulo, IME-USP
São Paulo, Brazil
davidbtadokoro@usp.br

Paulo Meirelles
University of São Paulo, IME-USP
São Paulo, Brazil
paulormm@ime.usp.br

***Abstract*—Large datasets of software code, such as Software Heritage and World of Code, assign authorship by commit and by line: the last author to change a line takes the whole line, and the data names persons, not the companies and foundations that maintain a Free/Libre and Open Source Software (FLOSS) project. We create Token on Me (ToM), a dataset that provides, for each token in the current code, the commit that added it and the organizations linked to its author and project, based on the author's e-mail domain and the repository namespace. ToM has 8,599 projects, selected from a pool of over 194,000 repositories, with at least 1,000 commits, and C, C++, Java, or Rust as the most common programming languages. Each row adds 21 columns to the usual git and cregit authorship columns: 17 for the project's provenance and 4 that link the author and the project to companies or organizations. ToM has over 31 billion rows, in one compressed Parquet file per project (116.9 GB).**

***Index Terms*—Mining Software Repositories, Free Software, Open Source, Code Authorship, Company Participation**

I. Introduction

Companies play a large part in the development of FLOSS [1]–[3]. In the Linux kernel releases 6.13 to 6.18 (2025), at least 86% of the 80,035 non-merge changesets come from developers with an employer or from consultants (5.3% have no employer and 8.5% have an unknown one) [4], and companies made about 90% of the commits to OpenStack [5].

Meanwhile, datasets of FLOSS authorship, such as World of Code [6] and Software Heritage [7], link authors to commits, files, and projects, not to the tokens of the current code. The most used form of authorship, `git blame`, works on lines: a change of one token, such as a renamed variable, moves the whole line to the author of that change [8]. Such limitations were tackled in disjoint ways by previous works: CREGIT [9] computes token-level blame in a repository, and Spinellis *et al.* [1] identify enterprise-driven GitHub projects using the e-mail domains of their commits. Moreover, Spinellis' approach did not provide the kind of organization[1] identified for each project, nor did it join it with authorship data.

To provide a unified solution to token-to-company authorship attribution, we present Token on Me (**ToM**), a token-level dataset of 8,599 FLOSS projects with 1,000 or more commits. For each token in the latest code, **ToM** provides, when applicable, the probable employer of its author and the organization that owns the project.

II. Related Work

Line-level authorship. Authorship and ownership studies commonly take the file or the line as their grain [10]–[12]. However, the limitations of overwriting the authorship attribution of every token of a changed line have also long been reported by MSR studies [13], [14]

Token-level authorship. CREGIT [9] turns each version of a file into a list of tokens, one token per line, and runs blame on that view, so each token keeps the commit that added it. Kondo et al. use CREGIT's tokens to study micro commits [14]. Qiu et al. use CREGIT to compare the committers of Linux kernel files and their employers with the copyright holders [15]. Our earlier study measured company authorship at token level in three Linux subsystems [16], with a replication package [17]. **ToM** applies our fork of CREGIT[2] to 8,599 projects.

Company participation and governance. Companies take part in FLOSS as employers of paid developers, as sponsors, and as owners of projects [18]–[20], and the governance of a community changes over time [21]. There is extensive published research on the dominance of companies in the OpenStack ecosystem [5], [22]; similar studies cover Android and four IoT standards [2], [3].

Related datasets. Software Heritage [7], World of Code [6] with its identity map [23], GHTorrent [24], and Boa [25] give commits, files, metadata, or AST queries at large scale, but no token authorship and no organization map beyond self-reported company fields and GitHub organization membership. Moreover, Cristutiu provides a commit-level dataset that links commits from 975 Python projects to organizations via e-mail domains [26].

[1]We use organization to refer to any body that employs developers or owns projects. A company is one kind of organization. Our dataset tells the kinds apart in its own tables and in the project columns.

[2]`github.com/ccsl-codev/cregit`

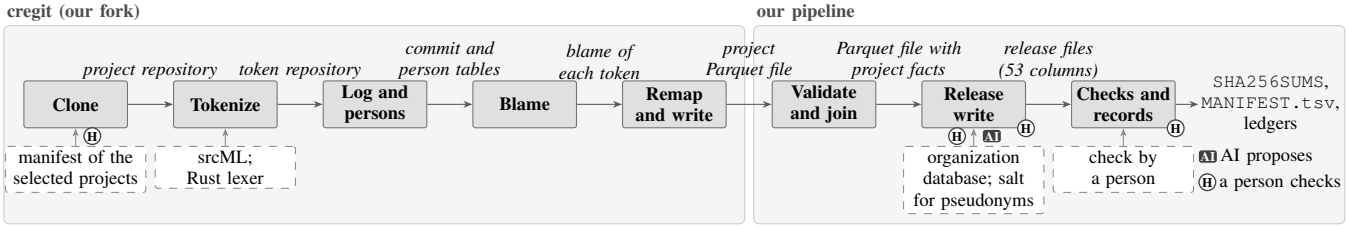


Fig. 1. Pipeline of **TOM**. The boxes are the steps, in run order. The italic text above each arrow is the data that a step gives to the next step. The dashed boxes are inputs from outside the run. An AI badge marks an input where AI proposes values; an H badge marks a human check.

III. Data Sources and Collection Strategy

We collected projects from five sources: after removing 8,992 duplicates, we ended up with a pool of 194,572 projects, as detailed below (total from each source in parentheses).

(a) **GitHub search:** repositories in all languages with more than 400 stars, not archived, not forks, and with a push on or after 2025-04-03 (80,990).
(b) **GitLab search:** projects with the same stars rule (75).
(c) **Spinellis *et al.*:** the repository URLs of the enterprise set, 17,264 projects [1], [27] (15,697).
(d) **Company organizations:** all public repositories of 506 GitHub organizations of 150 companies from the enterprise set (81,425).
(e) **Foundation and sponsor rosters:** the project lists of 24 foundations, 4 fiscal sponsors, and 2 companies from the list of open source organizations on opensource.com[3], and the group namespaces of 11 GitLab hosts and 22 gitlab.com groups, without filtering on stars (25,377).

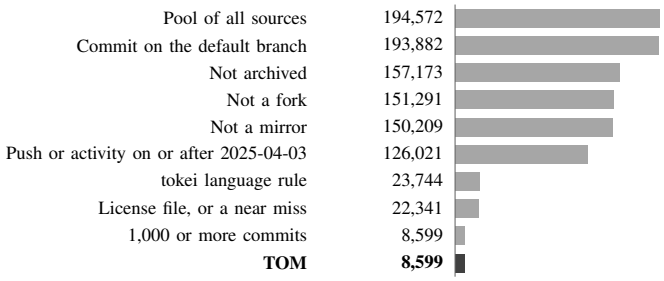


Fig. 2. Selection funnel of **TOM**.

We selected the projects from the pool that CREGIT and our pipeline can process. Our inclusion criteria for projects are (as per Figure 2): the default branch has a commit, and the project is not archived (both from the GitHub and GitLab metadata); it is not a fork (the fork flag of the host or on GitLab, also the same root commit as another pool project); it is not a mirror (the mirror flag of the host; on GitLab, also a project with the same name that holds the head commit of another pool project); it has a push or other activity on or after 2025-04-03 (the last 18 months); its largest language, calculated with TOKEI version 14.0.0, is C, C++, Java, or Rust; it has a license file (considering `LICENSE` or `LICENSE.md`, in any folder); and it has 1,000 or more commits.

TOM has 8,599 projects from GitHub search (6,905), GitLab search (11), Spinellis *et al.* [1] (805), the company organizations (591), and the foundation and sponsor rosters (1,573); a project can come from more than one source. Of these projects, 2,506 are C++, 2,242 are C, 2,215 are Java, and 1,636 are Rust. We suggest six size classes by commit count: **TOM** has 1,732 SS projects (fewer than 1,500 commits), 1,985 S (1,500 to 2,499), 2,058 M (2,500 to 4,999), 1,827 L (5,000 to 14,999), 719 XL (15,000 to 49,999) and 278 XXL (50,000 or more).

To support linking authorship to company data, we built an **organization database** through four steps: (i) an automatic script got the e-mail domains from public lists, such as the GITDM map of the Linux kernel, the CNCF GITDM affiliation list[4], company domains from Spinellis *et al.* [1], and the default domains of the foundation and sponsor rosters; (ii) we added to those domains the legal names, kinds, countries, related companies and organizations namespaces from Wikidata, the SEC, GLEIF and the GitHub organization pages; (iii) for the e-mail domains with no organization that hold the top 99% of the authored tokens, we ran one AI model (Claude Opus 5.5) in two roles, each with its own prompt and context: first it suggested a new or an existing organization based on the DNS resolution evidence of those domains, then, in a separate run, it judged the suggestion and checked the links and conflicts; (iv) last, there was a human audit of the items that the AI could not decide and of samples of the links to enable for a measured error rate (Section VI). The database has 9,340 organizations, 10,931 domain rows and 7,061 namespace rows.

Each project goes through the steps of Figure 1; the first five steps come from our fork of CREGIT [9][1]; our pipeline[5] adds the other steps. The runs took place from 2026-10-07 to 2026-11-10. The pipeline clones the default branch based on a MANIFEST.tsv generated by the selection step at its newest commit when the run starts and splits each source file into tokens, with srcML [28] for C, C++, and Java; and the `ra-ap-rustc_lexer` crate[6] for Rust. It reads the commits and assigns each author a person ID; CREGIT merges the aliases of a developer inside each project, as it groups the git

[3] `opensource.com/resources/organizations`

[4] `git.lwn.net/gitdm.git` and `github.com/cncf/gitdm`
[5] `github.com/ccsl-codev/cregit-token-pipeline`
[6] `crates.io/crates/ra-ap-rustc_lexer`

identities with the same name and joins the groups that share an e-mail address. It then finds the commit for each token using `git blame -C100 --line-porcelain` on the token view of each file. It maps that commit to the original repository, writes one Parquet file per project, and checks the rows and the schema of the file. At release, the pipeline pseudonymizes the authors, adds the project facts and the organization columns, and fills in empty language and license values. For each file, the release records the project commit, the tool versions, and a checksum, so a reader can verify the file and rebuild it.

IV. Storage and Schema

ToM has two parts (Figure 3): the project files and the organization database. Each project in **ToM** has one Apache Parquet file, with one row per token of the source code at the commit pinned for the run. The 8,599 Parquet files contain about 31.6 billion rows from 17.9 million source files and 437 thousand distinct person IDs: 116.9 GB in total, compressed with ZSTD. Each row has 53 columns in six groups (Table I).

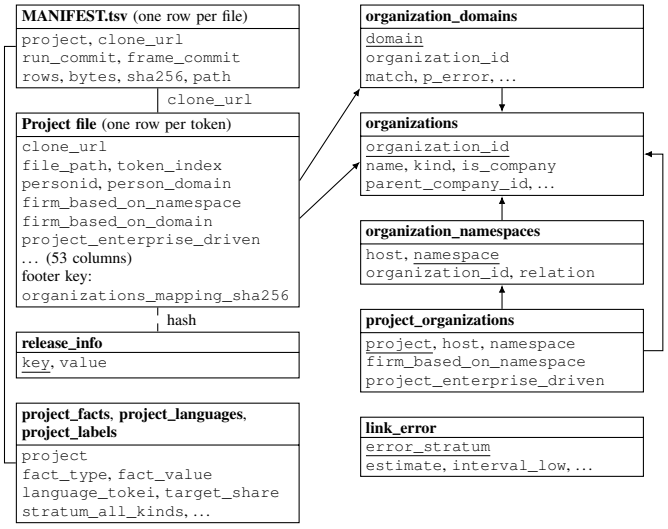


Fig. 3. `MANIFEST.tsv` and the project files hold the tokens; the other tables are the organization database. Keys are underlined. A foreign key is pointed to with an arrow. The joins of `MANIFEST.tsv`, by `clone_url` to the project files and by `project` to the project tables are straight lines. The footer hash of each file equals the hash in `release_info` is dashed.

Of all tokens, 56.8% get their domain link from the curated GITDM map and 26.2% from our database mappings done by the AI model; the other 17.0% from other lists and manual links (9.7%), or have no domain row and get `independent` or `unknown` by rule (7.3%).

The organization database has nine tables: (i) the organizations with their kind and parent; (ii) the maps from e-mail domains and from namespaces to organizations; (iii) the links of projects to organizations; (iv) three project tables; (v) the error estimates of the links; and (vi) the release information. The organization columns of **ToM** hold only IDs that point to the organizations table, and a hash in the footer of each project file indicates which version of the tables produced those IDs.

TABLE I
The 53 columns of each row, by group.

Group	#	Example columns	Values
Project	17	`owner`, `source`	from the selection step
		`stratum`, the governance type	`foundation`, `company`, `personal`, `contested`, `unknown`
		`license`	SPDX or `NOASSERTION`
Token	8	`file_path`, `token_value`	from CREGIT
Commit	8	`author_date`, `commit_summary`	from git
Person	2	`personid`	pseudonymized person
		`person_domain`	e-mail domain
Trailers	14	Assisted-by, Reviewed-by	pseudonymized persons
Organization	4	`firm_based_on_namespace`	owner of the project namespace
		`firm_based_on_domain`	organization of the e-mail domain
		`project_enterprise_driven`	our token version of the enterprise rule [1]
		`project_enterprise_basis`	tests that hold, *e.g.,* 2 of the 3 top authors share a company

A DuckDB file has views over the project files and the tables. The organization database can also be used on its own to map e-mail domains and namespaces to organizations.

V. Using the Dataset

ToM can be used to study company participation and answer questions about company authorship in FLOSS ecosystems. For instance, our preliminary study on firm-level knowledge concentration [16] can now cover all 8,599 projects. In **ToM**, whose pool intentionally includes company projects, company e-mail addresses hold 32.1% of the tokens, free e-mail providers 33.9%, and projects, independent developers, foundations, and other kinds 33.9%. The companies' share is largest in the XXL projects, at 47.5% of their tokens, and smallest in the SS projects, at 18.9%. Company work is also concentrated: of the 3,647 companies with tokens, the top 10 hold 45.7% of the company tokens.

For future research, **ToM** supports studies of how much FLOSS projects depend on companies and on individuals, across the governance labels; of what happens to the code when contributors leave a project [29]–[32]; of who writes the code against who reviews and accepts it, from the commit trailers [33]; of paid and volunteer work, measured by the code that survives [34], [35]; of corporate acquisitions [36]; of AI-assisted code, with the Assisted-by and Co-authored-by trailers as weak labels; of project survival [37]; and of the method itself, such as token blame against line blame.

VI. Final Remarks

A. Limitations

This dataset release includes only surviving code; code that was deleted or superseded at the moment of the pinned commit in MANIFEST.tsv is not included. We intend to extend **ToM** to include the history of all tokens in the near future. The

supported languages are C, C++, Java, and Rust; TypeScript, JavaScript, and Python are not included, even though they are also prominent languages used in FLOSS projects. When `git blame -C100` of a file took more than 3,600 s, we used plain blame, which does not follow code that moves between files: at least 547 files in 187 projects (0.003% of the files). Files that the tokenizer cannot parse in time are omitted; these total 92 files across 41 projects (0.0005% of the files). The `personid` is null in 97,950 rows of 42 projects, and bots are not filtered.

The domain of an e-mail address does not prove the employer: employees can commit their work with a personal e-mail address. A human audit of 514 mail domains estimates that 2.9% of tokens (95% interval: 1.5 to 4.9%) have an incorrect `firm_based_on_domain`; these errors can move the company share up or down, so the share is not a strict minimum. The same AI model proposed and judged the links in separate runs, and the human checker saw the source of each link.

B. Quality Checks Performed

Human checks (Figure 1 and Table II). The token files and the license values were collected without AI. The authors checked the output of the rules and the AI in three audits. (i) *Domain links.* A hand audit of 514 e-mail domains, sampled in strata with a chance in proportion to their tokens, found that 2.9% of the tokens have a wrong `firm_based_on_domain` (1.5 to 4.9%). (ii) *Namespace links.* An audit of 650 project namespaces, sampled in strata and weighted by their projects, found none wrong; because a few namespaces hold most of the sampled projects, the upper bound is 6.0% of the projects. (iii) *AI judge.* Seeded hand checks of 924 answers from three rounds of the AI judge found 8 wrong (0.9%, 0.4 to 1.7%). All intervals are 95% Clopper-Pearson intervals; for (i) and (ii), they use the effective sample size of each stratum.

TABLE II
AI AND RULE PROPOSALS AND HUMAN CHECKS.

Step	Proposed by	Human check	Result
Selection	rules, tokei	copy decisions	–
Token files	cregit, pipeline	one package rebuild	equal
Organizations	**AI** judge (web, DNS)	924 seeded answers	0.9% wrong*
Domain links	rules, **AI** judge	514 domains	2.9% of tokens wrong
Namespaces	rules	650 links	0 wrong

*the checker saw the AI verdict (lower bound)

File checks. All files have the same 53 columns, and the key columns have no nulls except in the 97,950 aforementioned rows (with null `personid`). A seeded sample of 31 projects has no duplicate keys. A rebuild of a reference project produces identical files, except for the pseudonyms.

External check on the Linux kernel. We compare the employer shares of the Linux kernel with the LWN.net employer statistics of the 7.3 cycle [38] (Table III). LWN counts only the changes of the cycle, while we count only the tokens alive at the 7.3 release. Over the employers that LWN lists, Spearman's ρ is 0.73 for our tokens against LWN's lines changed and 0.84 for our commits against LWN's changesets ($p < 0.01$ and $p < 0.0001$). The values differ because our dataset only looks at code at a pinned version, so deleted or overwritten code is not counted, and because LWN counts all files, while we count only C, C++, Java, and Rust files, without assembly, device trees, or scripts.

TABLE III
LINUX KERNEL EMPLOYER SHARES (%): OUR ALIVE TOKENS (ALL AND 7.3 CYCLE) AND SURVIVING 7.3 COMMITS, AGAINST LWN.NET LINES AND CHANGESETS OF THE 7.3 CYCLE.

	All	7.3 tokens		7.3 commits	
Employer	Ours	Ours	LWN	Ours	LWN
AMD	16.3	29.4	22.3	5.7	6.3
Intel	7.0	3.3	4.4	5.4	6.3
Red Hat	3.8	1.4	3.8	2.4	4.5
Linux Foundation	3.6	8.1	n.l.	12.6	n.l.
Qualcomm	3.0	6.4	6.1	3.6	5.9
Google	1.3	2.0	5.6	3.8	6.1
No employer	17.4	16.9	18.6	34.7	27.1
Spearman ρ		0.73 ($n = 17$)		0.84 ($n = 16$)	

C. Future Improvements

Later versions of the dataset can add smaller projects, more languages, the missing companies with a new manual human audit, a flag for vendored and generated code, a bot flag to see bot-authored code, and the dates of company mergers, acquisitions, and closures, so that a study can compare them with the commit dates. We also consider expanding the dataset to include deleted and superseded code, rather than just surviving tokens, thereby capturing the full history of every repository in the sample at the token level.

DATA AVAILABILITY

TOM, with the project files and the organization database, is public on the Internet Archive, archive.org/details/token-on-me-1.1.0; its Zenodo record, doi.org/10.5281/zenodo.24173700, gives the script that downloads and checks the data. A mirror is on Hugging Face, doi.org/10.57967/hf/11596. A standalone copy of the organization database is also archived on Zenodo [39], with the DOI doi.org/10.5281/zenodo.23744044. The replication package is on Zenodo, doi.org/10.5281/zenodo.24177100, and in the repository github.com/EllianCarlos/cregit-dataset-paper. It includes commands to check and rebuild the files and to list the tagged commits of the CREGIT fork and the pipeline. Software Heritage archives the code of all three. The data are under CC BY 4.0, and the token text keeps its project's license; the CREGIT fork and the pipeline are under GPLv3.0; the replication package is under Apache-2.0 (code) and CC BY 4.0 (documents and data).

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